

Week 6 Baseline Report: Emergency Triage Classification

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Decision supported

I tested whether simple classifiers can support a triage nurse by estimating the recorded Emergency Severity Index (ESI) level from information available at triage. The models do not assign care independently and are not ready for clinical use.

Logistic regression correctly identified only 4 of the 16 ESI 1 visits in the held-out test set.

Dataset recap

The supplied dataset contains 55,121 emergency department visits and 225 usable columns after removal of the exported index. The target has five levels, from ESI 1, which requires immediate intervention, to ESI 5. Only 77 visits are ESI 1, compared with 27,010 ESI 3 visits. That imbalance makes overall accuracy a poor safety measure on its own.

I used 209 numeric inputs, including age, front-door vital signs and chief-complaint flags. I excluded disposition and previousdispo because they are known after triage. Unencoded string fields were held out. The data was split 80/20 with stratification on ESI and a fixed random seed of 42. The test set contains 11,025 visits, including 16 ESI 1 cases.

Models

The comparison floor is a stratified dummy classifier. Logistic regression uses standardised inputs, with the scaler fitted on training data only. The decision tree uses unscaled values and is limited to a depth of five so the first tree remains readable and less likely to memorise the training set.

Benchmark results

Model	Accuracy	Macro F1	Weighted F1	ESI 1 recall
Stratified dummy	0.375	0.204	0.375	0.000
Logistic regression	0.683	0.508	0.677	0.250
Decision tree, depth 5	0.547	0.207	0.448	0.000

Logistic regression is the strongest baseline on every reported measure. The tree scores above the dummy on accuracy, but it predicts almost everything as ESI 3. The 0.169 gap between logistic regression's weighted and macro F1 shows how common classes can make the model look safer than its rare-class results.

Class-level results

Logistic regression

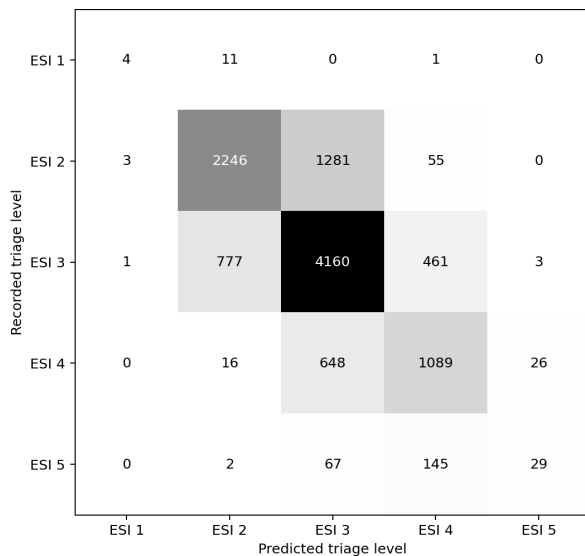
ESI	Precision	Recall	F1	Test visits
1	0.500	0.250	0.333	16
2	0.736	0.626	0.677	3,585
3	0.676	0.770	0.720	5,402
4	0.622	0.612	0.617	1,779
5	0.500	0.119	0.193	243

Decision tree, depth 5

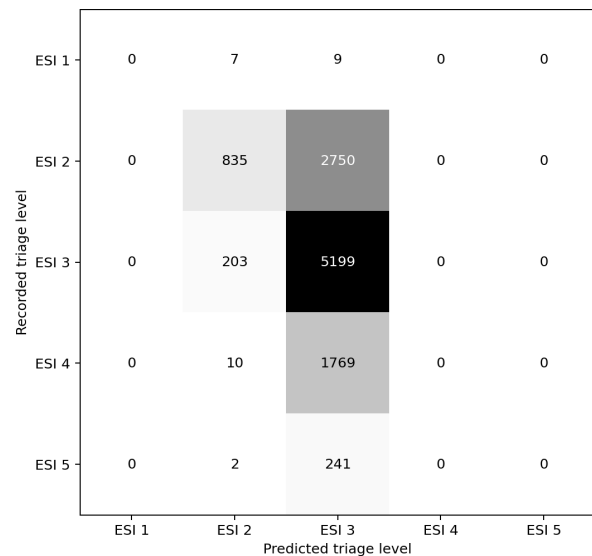
ESI	Precision	Recall	F1	Test visits
1	0.000	0.000	0.000	16
2	0.790	0.233	0.360	3,585
3	0.522	0.962	0.677	5,402
4	0.000	0.000	0.000	1,779
5	0.000	0.000	0.000	243

The tree never predicts ESI 1, 4 or 5 in the test set. Logistic regression is more balanced, but it detects only one quarter of ESI 1 cases and 11.9% of ESI 5 cases.

Logistic regression: test-set confusion matrix



Decision tree: test-set confusion matrix



Rows are the recorded ESI level and columns are the predicted level. Logistic regression misses 12 of 16 ESI 1 visits: 11 go to ESI 2 and one goes to ESI 4.

Primary metric and clinical failure

I chose ESI 1 recall as the primary safety measure because it answers a direct clinical question: of the patients who need immediate intervention, how many does the model identify at that level? Logistic regression finds 4 of 16, for recall of 0.25. It misses 12. A false negative could place a patient who needs immediate resuscitation into a slower pathway.

The failure is plausible given the data. ESI 1 has only 77 examples in the full dataset, and the first baseline uses numeric triage fields without the unencoded arrival and demographic fields. The target is the recorded ESI decision rather than an independently adjudicated clinical outcome. This run cannot show whether each miss reflects scarce examples, an ambiguous presentation, label noise or a missing signal.

The result rules out deployment. Although ESI 2 is still urgent, the model has not shown that it can separate immediate from emergent care with acceptable consistency. Accuracy does not offset that failure.

Next experiment

The next experiment should test class weighting and a one-versus-rest probability threshold for ESI 1 on the training data only. I would compare any gain in ESI 1 detection with the number of false alarms, because an alert that fires too often can lose clinical trust. Cross-validation, probability calibration and subgroup checks should follow before an external dataset or prospective workflow test. A clinician should review the ESI 1 errors before any model change is treated as an improvement.

Limits

This is a retrospective result from one supplied dataset. Categorical fields were held out, the target may contain labelling variation, and the models have not been calibrated, externally validated or checked prospectively. No subgroup fairness claim is made. The models remain exploratory decision-support baselines.

Reproducibility and submitted artefacts

Item	Recorded value
Random seed	42
Split	80/20, stratified on esi
Models	Stratified dummy; scaled logistic regression; decision tree, max_depth=5
Notebook	notebooks/week6_final_baseline_models.ipynb
Metrics	docs/w6_final_metrics.csv; docs/w6_per_class_metrics.csv
Figures	docs/w6_confusion_logreg.png; docs/w6_confusion_tree.png
Explainer	docs/week6_clinical_explainer.mp4 and transcript

The full clinical CSV remains outside GitHub.